



JKBOSE 2026

CLASS 11TH

PERMUTATION AND COMBINATIONS

MATHS

LECTURE- 06

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TOPICS TO BE COVERED

1

Properties of n_{C_r}

$$n(n-1)! = n!$$

$$(n+1-r)! = (n+1-r)(n+1-r-1)! \\ = (n-r+1)(n-r)!$$

2

Exercise Discussion 6.4

$$n_{C_{r-1}} = \frac{n!}{(r-1)!(n-(r-1))!} = \frac{n!}{(r-1)!(n-r+1)!}$$

3

Related Problems

$$n+1 = k(n-r+1)! = (n-r+1)(n-r)! \\ \frac{(n+1)!}{r!(n+1-r)!} = \frac{k!}{r!(k-r)!} = k_{C_r} = (n+1)_{C_r}$$

Properties of nCr

$$* nCr = nC_{n-r}$$

Proof: Here $nCr = \frac{n!}{(n-r)!r!}$ — (1)

$$\& nC_{n-r} = \frac{n!}{(\cancel{n-n+r})!(n-r)!} = \frac{n!}{(n-r)!r!} \text{ — (2)}$$

From (1) & (2), we get;

$$nCr = nC_{n-r}$$

Hence proved.

$$* nCx = nCy \Rightarrow x=y \text{ or } x+y=n$$

Proof: we know that

$$nCy = nC_{n-y}$$

$$\Rightarrow nCx = nCy = nC_{n-y}$$

$$\Rightarrow x=y \text{ or } x=n-y$$

$$\Rightarrow x=y \text{ or } x+y=n$$

Hence proved.

* $n C_r + n C_{r-1} = {}^{n+1}C_r$ (Pascal's rule)

Proof: we have

$$n C_r + n C_{r-1} = \frac{n!}{r!(n-r)!} + \frac{n!}{(r-1)!(n-r+1)!}$$

$$\Rightarrow n C_r + n C_{r-1} = \frac{n!}{r(r-1)!(n-r)!} + \frac{n!}{(r-1)!(n-r+1)(n-r)!}$$

$$\Rightarrow n C_r + n C_{r-1} = \frac{n!}{(r-1)!(n-r)!} \left\{ \frac{1}{r} + \frac{1}{n-r+1} \right\}$$

$$\Rightarrow n C_r + n C_{r-1} = \frac{n!}{(r-1)!(n-r)!} \left\{ \frac{\cancel{n-r+1} + \cancel{r}}{(n-r+1)r} \right\}$$

$$\Rightarrow n C_r + n C_{r-1} = \frac{(n+1)n!}{(r-1)!(n-r)!(n-r+1)r}$$

$$\Rightarrow n C_r + n C_{r-1} = \frac{(n+1)!}{r!(n+1-r)(n+1-r+1)!}$$

$$\Rightarrow n C_r + n C_{r-1} = \frac{(n+1)!}{r!(n+1-r)!} = {}^{n+1}C_r$$

$$\Rightarrow n C_r + n C_{r-1} = {}^{n+1}C_r \quad \text{Hence proved}$$

$$* \frac{{}^nC_{r-1}}{{}^{(n-1)}C_{r-1}} = \frac{n}{n-r+1}$$

Proof: we know that

$${}^nC_{r-1} = \frac{n!}{(r-1)!(n-r+1)!} \quad \text{--- (1)}$$

$$r \cdot {}^{(n-1)}C_{r-1} = \frac{(n-1)!}{(r-1)!(n-r+1)!} \quad \text{--- (2)}$$

Dividing (1) & (2), we get,

$$\frac{{}^nC_r}{{}^{(n-1)}C_{r-1}} = \frac{n!}{\cancel{(r-1)!} (n-r+1)!} \times \frac{\cancel{(r-1)!} (n-r)!}{(n-1)!}$$

$$\Rightarrow \frac{{}^nC_r}{{}^{n-1}C_{r-1}} = \frac{n \cancel{(n-1)!}}{(n-r+1) \cancel{(n-r)!}} \times \frac{\cancel{(n-r)!}}{\cancel{(n-1)!}}$$

$$\Rightarrow \boxed{\frac{{}^nC_r}{{}^{n-1}C_{r-1}} = \frac{n}{n-r+1}}$$

Hence proved.

Properties of nC_r



$$n-1-(r-1) = n-1-r+1 = n-r$$

- $nC_r = nC_{n-r}$
- $nC_x = nC_y$ and $x \neq y \Rightarrow x + y = n$
- $nC_r + nC_{r-1} = (n+1)C_r$ (**Pascal's Rule**)
- $\frac{nC_{r-1}}{(n-1)C_{r-1}} = \frac{n}{(n-r+1)}$ **
- $\frac{nC_r}{(n-1)C_{r-1}} = \frac{n}{r}$ ***
- $\frac{nC_r}{nC_{r+1}} = \frac{r+1}{n-r}$ *** (H/W)

→ **Proof:** we know that

$$nC_r = \frac{n!}{r!(n-r)!} = \frac{n(n-1)!}{r(r-1)!(n-r)!}$$

$$\Rightarrow nC_r = \frac{n}{r} \times \frac{(n-1)!}{(r-1)!(n-1-(r-1))!}$$

$$\Rightarrow nC_r = \frac{n}{r} \times {}^{n-1}C_{r-1}$$

hence proved.

EXAMPLE

H/W

If $n_{C_{r-1}} : n_{C_r} : n_{C_{r+1}} = 3 : 4 : 5$, then find the values of n and r .

QUESTION

A student needs to choose 3 books to read from a list of 10. In how many ways can the books be chosen?

1

1000 X

$${}^{10}C_3 = \frac{10!}{3!7!} = \frac{10 \times 9 \times 8 \times \cancel{7!}}{3! \cdot \cancel{7!}}$$

2

120 ✓

$$= \frac{10 \times \cancel{3} \times \cancel{8}}{\cancel{3} \times \cancel{2} \times 1} = 10 \times 12 = 120.$$

3

30 X

4

720 X

QUESTION

A committee of **5 people** is to be **formed** from a group of **6 men and 4 women**. How many **committees can be formed** if there are **no restrictions**?

- 1 252 ✓
 - 2 30240 ✗
 - 3 120 ✗
 - 4 210 ✗
- $$\begin{aligned}
 & {}^{10}C_5 \\
 &= \frac{10!}{5!5!} = \frac{10 \times 9 \times 8 \times 7 \times 6 \times 5!}{120 \times 5!} \\
 &= \frac{2 \times 3 \times 4 \times 7}{2 \times 5} = 252
 \end{aligned}$$

QUESTION

In how many ways can a cricket team of 11 players be chosen from 15 players if a particular player is always included?

1

1365 X

2

1001 ✓

3

110 X

4

364 X

$$\begin{aligned}
 & {}^1C_1 \times {}^{14}C_{10} \\
 & = {}^{14}C_{10} = \frac{14!}{10!4!} = \frac{14 \times 13 \times 12 \times 11 \times 10!}{10! \times 24} \\
 & = 77 \times 13 \\
 & = 1001
 \end{aligned}$$

$$\begin{array}{r}
 77 \times 13 \\
 \hline
 231 \\
 77 \times \\
 \hline
 1001
 \end{array}$$

QUESTION

A bag contains **5 red, 4 blue, and 3 green marbles**. In how many ways can **3 marbles** be chosen such that **at least one is green**?

1

36 ✗

2

220 ✗

3

136 ✓

4

84 ✗

(Sol) Required No. of Selections

$$= {}^3C_1 \times {}^9C_2 + {}^3C_2 \times {}^9C_1 + {}^3C_3 \times {}^9C_0$$

$$= 3 \times \frac{9 \times 8 \times 7}{7! \times 2!} + (3 \times 9) + (1 \times 1)$$

$$= (3 \times 36) + 27 + 1$$

$$= 108 + 28 = 136$$

EXERCISE – 6.4

QUESTION

If ${}^nC_8 = {}^nC_2$, find nC_2 .

(Sol) $n = 8 + 2 = 10$

$${}^nC_2 = {}^{10}C_2 = \frac{10 \times 9}{2} = 45.$$

H/W.

QUESTION

Determine n if

(i) ${}^{2n}C_3 : {}^nC_3 = 12 : 1$

(ii) ${}^{2n}C_3 : {}^nC_3 = 11 : 1$ (H/W)

(Sol) (i) Given that ${}^{2n}C_3 : {}^nC_3 = 12 : 1$

$$\Rightarrow \frac{(2n)!}{(2n-3)! \cdot 3!} : \frac{n!}{(n-3)! \cdot 3!} = 12 : 1$$

$$\Rightarrow \frac{(2n)!}{(2n-3)! \cdot 3!} \times \frac{(n-3)! \cdot 3!}{n!} = \frac{12}{1}$$

$$\Rightarrow \frac{2n(2n-1)(2n-2)(2n-3)!}{(2n-3)!} \times \frac{(n-3)!}{n(n-1)(n-2)(n-3)!} = \frac{6}{1}$$

$$\Rightarrow \frac{(2n-1)(2n-2)}{(n-1)(n-2)} = 6$$

$$\Rightarrow \frac{2(2n-1)(n-1)}{(n-1)(n-2)} = 6$$

$$\Rightarrow \frac{2n-1}{n-2} = 3$$

$$\Rightarrow 2n-1 = 3n-6$$

$$\Rightarrow 2n-3n = -6+1$$

$$\Rightarrow -n = -5$$

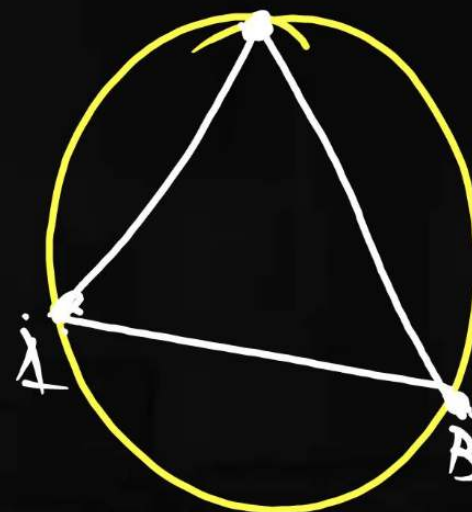
$$\Rightarrow \boxed{n=5}$$

(Hence the result)

QUESTION

How many chords can be drawn through 21 points on a circle?

$${}^{21}C_2$$



QUESTION

In how many ways can a team of 3 boys and 3 girls be selected from 5 boys and 4 girls?

$${}^nC_r = \frac{n!}{r!(n-r)!}$$

$${}^5C_3 \times {}^4C_3$$

Required No. of selections

$$= {}^5C_3 \times {}^4C_3$$

QUESTION

Find the number of ways of selecting 9 balls from 6 red balls, 5 white balls and 5 blue balls if each selection consists of 3 balls of each colour.

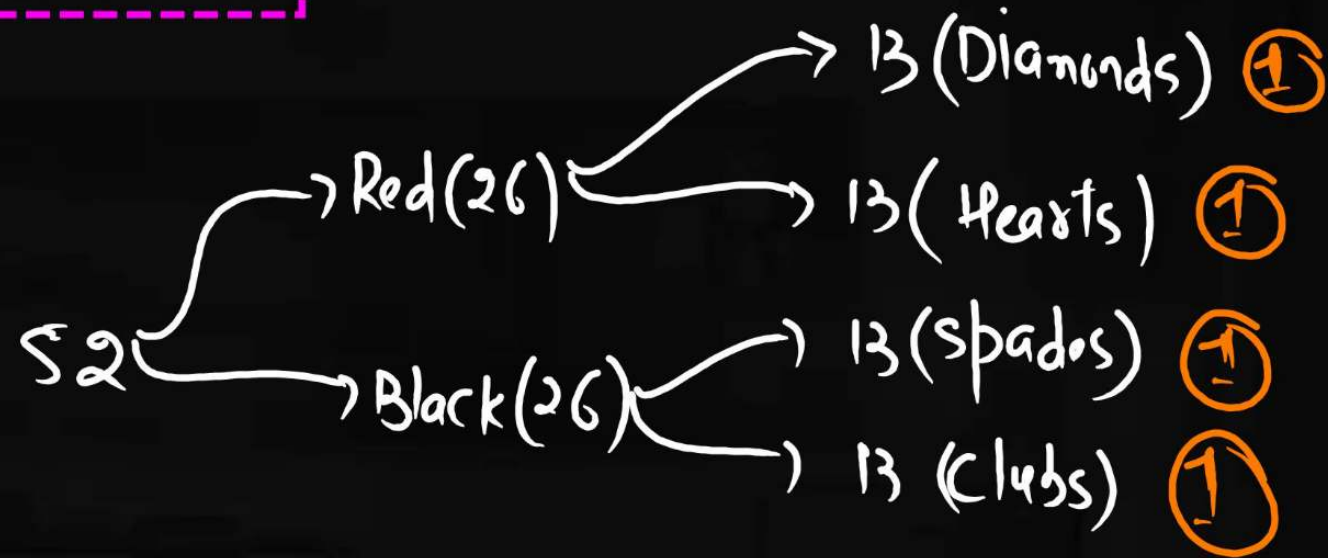
$${}^6C_3 \times {}^5C_3 \times {}^5C_3$$

QUESTION

Determine the number of 5 card combinations out of a deck of 52 cards if there is exactly one ace in each combination.

$${}^4C_1 \times {}^{48}C_4$$

$${}^4C_1 \times {}^{48}C_4$$



48

QUESTION

In how many ways can one select a cricket team of eleven from 17 players in which only 5 players can bowl if each cricket team of 11 must include exactly 4 bowlers?

$${}^5C_4 \times {}^{12}C_7$$

17 $\begin{cases} \rightarrow 5 \text{ Bowler} \\ \rightarrow 12 \text{ Batters} \end{cases}$

QUESTION

A bag contains 5 black and 6 red balls. Determine the number of ways in which 2 black and 3 red balls can be selected.

$${}^5C_2 \times {}^6C_3$$

QUESTION

In how many ways can a student choose a programme of 5 courses if 9 courses are available and 2 specific courses are compulsory for every student?

$$= {}^2C_2 \times {}^7C_3$$

$$= {}^7C_3$$

9 $\left\{ \begin{array}{l} 2 \text{ compulsory} \\ 7 \text{ optional} \end{array} \right.$

Thank
You